SpeedLimit

Speed recommendations from weather and air quality

Mario Palagyi & Jerome Puthuparampil

Local speed limit setting website prototype that lets operators automatically or manually adjust speed limit based on weather and air conditions

Project and approach

Stakeholders

Drivers and passengers, road operators, traffic authorities, emergency services, nearby residents, environmental agencies, data and sensor experts, software engineers, and legal specialists.

Three-step exercise process

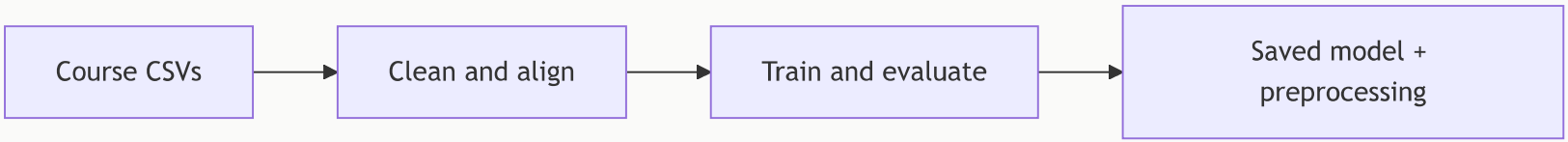
We completed the exercises through manual work, an initial AI prompt, and a refined prompt, following the process defined in class.

AI use and lessons learned

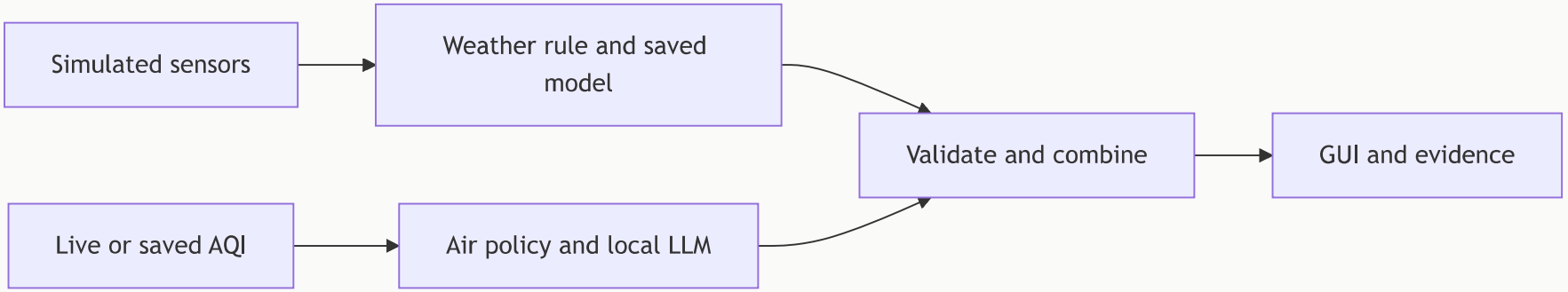
- We mainly present the initial-prompt outputs because we judged them the most useful for this prototype.
- With GPT 6.1 Sol, more elaborate prompts sometimes added complexity without improving the result.
- Some manual and initial-prompt solutions were similar. We kept the classroom prototype simple where more complexity didn't offer any benefit.

Training and operation

Training: clean historical records, align hours, train and save the weather model.



Production: different steps depending on weather or air branch, validate and combine, then decision in GUI



Weather branch invokes only on trigger of condition · Air runs independently

03 / 07

DATA ENGINEERING

Data quality shaped the model

405 / 744

usable Year N hours

Inputs at hour start: light, water, temperature, recorded speed limit.

Target: near-accidents in the following hour.

Sensor meaning

Resolve ambiguous sensor identities.

IDs 15/18 provisionally represent temperature. ID 16 is ambiguous; ID 19 has no inventory entry. Both are excluded from model inputs.

Event timing

Set aside events with unresolved times.

44 events use hour 24, including event-file row 74. Incidents use hours 1–24; sensors use 0–23. The full hour convention remains unresolved.

Time alignment

Use past readings; exclude speed-changing hours.

A 09:50 reading can supply the 10:00 example. One speed per hour simplifies reality and loses data. Further engineering could model shorter intervals after timestamp clarification.

> Provisional assumptions: temperature IDs, event-hour interpretation, and complete event logs for zero-event hours.

EVIDENCE

Demonstration and measured results

Clear	Black Ice	Dark	Poor air forecast	Both
80 km/h	**70** km/h	**40** km/h	**50** km/h	**40** km/h

64 automated tests passed and five recorded real-model checks passed.

Evaluation dataset	Network MAE	Baseline MAE
Year N test	**0.86**	0.91
Year N+1	0.91	**0.89**

MAE: average absolute prediction error, measured in near-accidents per hourly example. Lower = better

Validation also favored the baseline. Predictive improvement remains limited.

Real-model checks and demos recorded 13 Nov

05 / 07

RISKS AND LESSONS

Limits that determine the next step

Limitation	Consequence
Data Engineering	Low predictive performance and oversimplified data engineering, leading to information loss, highlighting this step’s importance.
No training hours at 40 km/h	The dark-road fallback has weak support. Every candidate misses the target in that example.
Year N+1 repeats sensor readings	Following-year results provide limited independent evidence.
LLM made numerical policy errors	Python supplies the checked peak and applicable rule, then validates the response.

> **Historical associations do not establish the safety effect of changing speed limit.**

The AQI-to-speed mapping is an educated assumption. Validation further needed

Target unmet and unsupported-speed warnings remain visible in the GUI

RECOMMENDATION

Continue controlled validation

Clarify data definitions

Confirm sensors (e.g. T, and IDs), event times, and logging coverage (0 accidents recorded hours).

Strengthen the model evidence

Collect observations at reduced speeds (40 km/h) and compare simpler models.

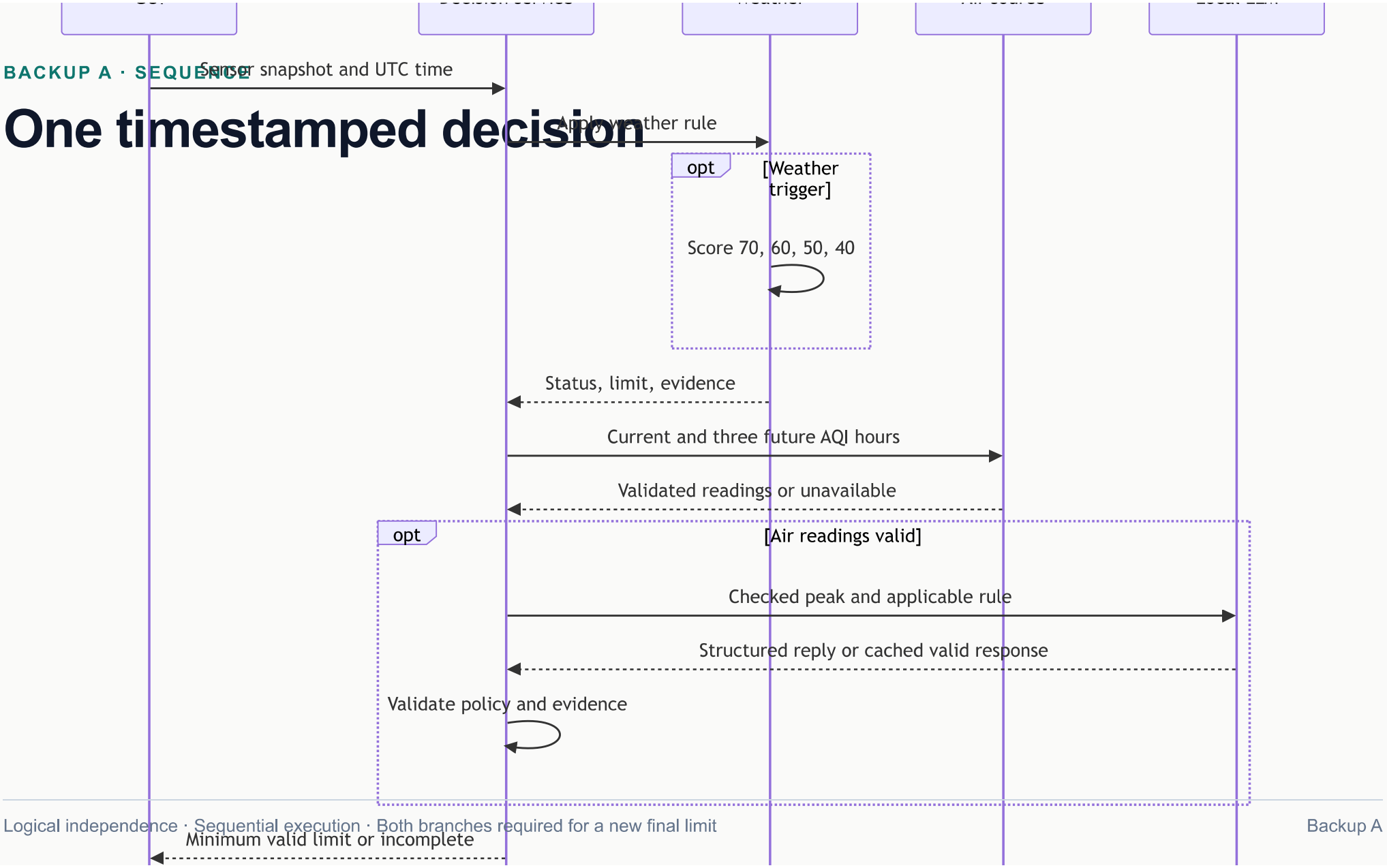
Validate the operating policy

Human oversight and failure response, then check reliability over time.

Further plans include: Examine data in detail, fix excluded data thus better model performance, human oversight

BACKUP A · SEQUENCE

One timestamped decision



Logical independence · Sequential execution · Both branches required for a new final limit

Backup A

BACKUP B · DATA AUDIT

Source-data findings and handling

Finding	Handling
Semicolons, decimal commas, water code `W`	Explicit parsing, numeric normalization, trimmed codes
Headerless N+1 reading file	Assign known columns and preserve the first observation
`T` means temperature and traffic	Use IDs 15/18 provisionally and exclude ambiguous ID 16
Unknown ID 19 and missing IDs 7/10/17	Exclude unknown readings and record inventory discrepancies
Hour-24 events: 44 in N, 31 in N+1	Quarantine invalid dates and plausible affected midnights
No event-coverage field	Assume complete logs except known uncertain hours
Identifying licence plates	Drop at the ingestion boundary
N+1 readings repeat N exactly	Flag limited independence in the assessment

Actual checked sensor readings had zero exact duplicates, invalid dates, missing numeric IDs/values, or out-of-range values.

Provisional mappings and event-hour semantics still require confirmation

Backup B

BACKUP C · FEATURE ENGINEERING

What one training row means

At 10:00: use weather and posted speed known by 10:00.
Target: near-accidents from 10:00 inclusive to 11:00 exclusive.

Step	Implemented rule
Align readings	Latest value at or before the boundary, at most two hours old
Combine sensors	Median across available sensors of the same feature
Normalize units	Lux multiplied by 1,000 once, water in µm, temperature in °C
Count events	One row counts once if any valid near-event measurement exists
Distinguish zero and unknown	Zero requires assumed coverage, unknown coverage excludes the hour
Keep usable targets	All features present, coverage acceptable, no within-hour speed change

> A 09:50 reading can describe 10:00. A 10:10 reading cannot. The first hour has no past boundary features.

405 usable N hours and 404 N+1 hours · Exclusion categories overlap

Backup C

BACKUP D · WEATHER MODEL

Model and evaluation

StandardScaler + MLPRegressor: four inputs, 16 ReLU hidden units, one count estimate.
Chronological 70/15/15 split, training-only scaling, seed 42.

Partition	Hours	Network MAE	Training-mean baseline MAE
N training	283	0.902	0.993
N validation	61	1.190	1.104
N test	61	0.860	0.910
N+1, frozen model	404	0.907	0.888

Training support at 40 / 50 / 60 / 70 / 80: 0 / 7 / 84 / 15 / 177 hours.

> **N+1 verdict: insufficient evidence.** Its sensor inputs repeat N exactly. The original model remains unchanged.

MAE measures average absolute count error in events/hour · No causal intervention test

Backup D

BACKUP E · WEATHER TESTS

Strict boundaries and fallback

```
light < 500 OR (water > 1000 AND temperature < 0)
```

Units: millilux, µm, °C. Clear weather returns 80 without a network call.

Light	Water	Temp.	Trigger
499	0	10	Yes
500	1000	-1	No
500	1001	-1	Yes
500	1001	0	No

Dark-road candidate	Estimated events/h
70 km/h	2.431
60 km/h	2.224
50 km/h	1.830
40 km/h	2.332

> Choose the highest candidate with estimate ≤1. If none qualifies, choose 40 and display **target unmet.**

The fallback chooses the lowest allowed speed · Scores can be nonmonotonic

Backup E

BACKUP F · AIR-QUALITY BRANCH

Air policy and guarded local inference

Take the maximum European AQI across the current reading and next three complete hourly forecasts.

Peak AQI	Reduction from 80	Air limit
0 through 60	0 km/h	80 km/h
Above 60 through 80	10 km/h	70 km/h
Above 80 through 100	20 km/h	60 km/h
Above 100	30 km/h	50 km/h

Python: verifies peak and selects the applicable rule.
Qwen: returns structured fields that Python validates against the rule and evidence.

> The GUI derives its sentence from checked fields. It retains original model prose and raw JSON for inspection.

Source: Open-Meteo / CAMS ENSEMBLE · Speed mapping: student assumption

Backup F

BACKUP G · OPERATIONAL BEHAVIOR

Reliability and failure behavior

Situation	Implemented behavior
Both branches valid	Final limit is the minimum, such as min(60,70) = 60
Required branch unavailable	Incomplete, no new final limit, last complete time remains visible
Weather target unmet	Lowest candidate with warnings, branch can still be valid
Start / Stop	Immediate decision then due ticks / stop future ticks
Slow request or GUI redraw	Busy guard avoids overlap, redraw alone causes no decision
API freshness	Cache up to 15 minutes, validate again and fail on unsuccessful expired refresh
Repeated identical LLM input	Reuse a previously validated response

Scheduling depends on the active page. Long unattended use and browser sleep/resume still need checks.

BACKUP H · MAINTAINABILITY

Architectural change requests

Proposed change	Components affected	Required work
Sensors report Fahrenheit	Input adapters and cleaning	Convert to Celsius before routing or inference, test boundary equivalence
Only skidding counts	Event target construction, training, evaluation	Rebuild labels in both years, retrain, compare results
Electric cars exempt from air restrictions	Vehicle inputs, air policy, combination, display	Preserve weather limits and resolve how vehicle-specific limits reach drivers

> Separate responsibilities make change impact easier to identify. These changes remain proposals.

Current scope: one laptop application, simulated weather inputs, a local weather model, Open-Meteo, and local Ollama.

Core modules: data, weather, air, decision orchestration, GUI

Backup H

BACKUP I · ACCOUNTABILITY

AI use and human oversight

Development and operation

A coding assistant wrote the implementation and tests.

The local Qwen model participates in the runtime air branch.

Prompt failures and revisions are recorded. The course's required AI-free comparison still needs its own evidence.

Decisions requiring people

Data owners: confirm sensor and event definitions.

Traffic specialists: approve the speed policy and failure procedure.

Engineers and operators: verify reliability and investigate uncertain results.

> Safety benefit, deployment readiness, and formal compliance remain unestablished.

BACKUP J · HOUR EXCLUSIONS

How 744 hours became 405

One example pairs conditions known at hour start with near-accidents during the following hour.

Preparation step	Newly excluded	Remaining
January: 31 days × 24 hours	—	744
Require readings available at hour start	1	743
Require interpretable incident coverage	29	714
Exclude hours with changing posted limits	309	**405**

There are 329 speed-changing hours overall. Twenty already fail the incident-coverage rule, so only 309 are additional exclusions. The 405 examples are then split into training, validation, and test sets.

Why exclude changing speeds? An hour with 80 km/h until 10:20 and 60 km/h afterwards cannot be faithfully represented by one speed for the entire hour.

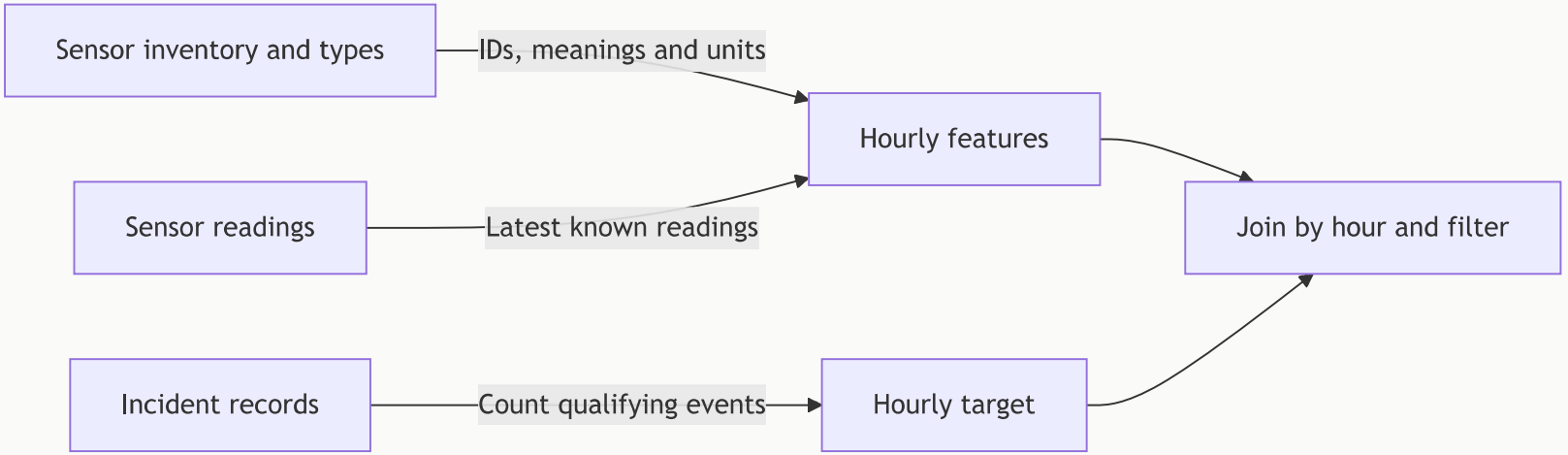
This is a simplifying model choice, not an unavoidable cleaning rule. It loses data and may introduce selection bias. Clarifying event times could enable shorter constant-speed intervals with duration-aware targets.

Counts reflect current preparation rules · Timestamp assumptions still need confirmation

Backup J

BACKUP K · DATA RELATIONSHIPS

How the source data connect



Sensor mapping: the readings' Sensor field matches inventory SensorID. The inventory's SensorTypeCode connects to the type definitions.

This mapping needs exceptions: T means both temperature and traffic, sensor 19 lacks an inventory entry, and special sensor −1 records the posted speed limit.

Incident relationship: events have no sensor-ID link. They connect to sensor-derived features through the constructed hourly timeline.

Incidents use hours 1–24 and sensors use 0–23. If incidents use an hour-ending convention, all event hours may need realignment. Quarantining hour 24 alone does not settle this question.

Four feature values and one incident-count target per eligible hour

Backup K

BACKUP L · FEATURE SELECTION

Model inputs and excluded fields

Weather-model input	Role
Light, in millilux	Represents lighting conditions
Road water, in µm	Represents water on the road
Temperature, in °C	Represents temperature and freezing conditions
Posted speed limit, in km/h	Allows predictions at different candidate limits

Target: qualifying near-accidents in the following hour. The data contain no labels identifying the optimal speed.

Other channels: humidity, wind, noise, and air pressure are outside the selected prototype feature set. This does not establish that they are irrelevant. Traffic cannot be confidently identified from the duplicated T code.

Unclear sensors: ID 16 has ambiguous meaning and ID 19 lacks an inventory entry. Temperature IDs 15/18 remain provisional.

Event fields: close-car, guardrail, and skid measurements construct the target rather than becoming inputs. Damage or injury alone does not qualify for this target. Licence plates are discarded.

Separate or unavailable information: AQI belongs to the air branch. Future readings and incident outcomes cannot be prediction inputs because they would leak future information.